AMAM Note

1 Notation & Perturbation Method

Big $\mathcal{O}$: $f(z)=\mathcal{O}(g(z))$ as $z\to z_0$ if $\lim\limits_{z\to z_0}\frac{f(z)}{g(z)}$ **bdd** $\Leftrightarrow |f(z)|\le A|g(z)|$ as $z\to z_0$

Small o: $f(z)=o(g(z))$ as $z\to z_0$ if $\lim\limits_{z\to z_0}\frac{f(z)}{g(z)}=0 \Leftrightarrow f(z)\ll g(z)$ as $z\to z_0$

Asymptotic $\sim$: $f(z)\sim g(z)$ as $z\to z_0$ if $\lim\limits_{z\to z_0}\frac{f(z)}{g(z)}=1$ **ps:** $e^{-x}=o(x^{-n}), \forall n>0$ **as** $x\to\infty$

Asymptotic Sequence: Seq $\{\varphi_n(z)\}$ is asy. if $\forall n\in\mathbb{N}\ \varphi_{n+1}(z)=o(\varphi_n(z))$ as $z\to z_0$

Asymptotic Expansion: $f(z)\sim\sum\limits_{n=0}^{M}a_n\varphi_n(z)$ as $z\to z_0$ to M if $\phi_n(z)$ **is asy. seq.** and

$f(z)-\sum\limits_{n=0}^{N}a_n\varphi_n(z)=o(\varphi_N(z))=\mathcal{O}(\varphi_{N+1}(z)), \forall N\le M$ as $z\to z_0$

$f(x,\varepsilon)\sim\sum\limits_{n=0}^{\infty}a_n(x)\varepsilon^n$ uniformly in $x\in I$ as $\varepsilon\to 0$ if ε 与 x 无关(i.e. $\sup\limits_{x\in I}|f(x,\varepsilon)-\sum\limits_{n=0}^{N}a_n(x)\varepsilon^n|=o(\varepsilon^N)$)

Singular|Regular - Perturbation Method: **a.** Let $\varepsilon=0$ solve $f(x,0)=0$.

b.i. 如果 f 结构变了(阶数降低), Let $x=\varepsilon^{-\alpha}X$, PDB 找 α **c.** 假设 $X=a_0+a_1\varepsilon^{\beta}+a_2\varepsilon^{2\beta}+\cdots$ 代入求解

b.ii. 如果 f 结构不变/变, 但有重根 x_0, Let $x=x_0+\varepsilon^{\alpha}X$, PDB 找 α. $\uparrow\beta$ 看化简后的 X 方程得出

Principle of dominant balance (PDB): For Singular Perturbation $p(x,\varepsilon)=0$, Let $x=\varepsilon^{\alpha}X$, 代入选择 α:

对于 ε 来说 **a.** 至少有两项阶数相同且最低; **b.** 其他项阶数均更高. (即可忽略)

2 Integral By Parts, Laplace's Method & Method of Steepest Descents

Taylor: $f(x)=\sum\limits_{n=0}^{N-1}\frac{f^{(n)}(x_0)}{n!}(x-x_0)^n+\frac{f^{(N)}(\xi)}{N!}(x-x_0)^N \Rightarrow f(x)\sim\sum\limits_{n=0}^{\infty}\frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$ as $x\to x_0$

Exp. Integral: $E_1(x)=\int_x^{\infty}\frac{e^{-t}}{t}dt$, ($x\to 0$ 发散) $E_1(x)\sim e^{-x}\sum\limits_{n=0}^{\infty}(-1)^n\frac{n!}{x^{n+1}}, x\to\infty$

Gamma Func: $\Gamma(x)=\int_0^{\infty}t^{x-1}e^{-t}dt$ $\Gamma(x+1)=x\Gamma(x)$ and $\Gamma(n)=(n-1)!$ $\Gamma(\frac{1}{2})=\sqrt{\pi}$

Incomplete Gamma Func: **Lower**: $\gamma(a,x)=\int_0^x t^{a-1}e^{-t}dt$ **Upper**: $\Gamma(a,x)=\int_x^{\infty}t^{a-1}e^{-t}dt$

- $\gamma(a,x)=\Gamma(a)-\Gamma(a,x)$ $\Gamma(a,x)\sim e^{-x}x^{a-1}\sum\limits_{n=0}^{\infty}\frac{\Gamma(a)}{\Gamma(a-n)}x^{-n}, x\to\infty$

Laplace's Method: For $I(x)=\int_a^b e^{xf(t)}g(t)dt$ as $x\to\infty$

1. If $f(t)$ has unique max at $t=t_0\in(a,b)\Rightarrow I\sim\int_{\infty}^{\infty}e^{x[f(t_0)+\frac{f''(t_0)}{2}(t-t_0)^2+\cdots]}g(t_0)dt$
   (留 n 项非常数展开为 n term approximation) (ps: max $\leftrightarrow f'(t_0)=0, f''(t_0)<0$)

   Useful Tool: Now: $I\sim e^{xf(t_0)}\int_{\infty}^{\infty}e^{x[\frac{f''(t_0)}{2}(t-t_0)^2+\cdots]}g(t_0)dt \Rightarrow$ Let $\tau^2=-[\frac{1}{2}f''(t_0)(t-t_0)^2+\cdots]$
   Assume $t=\tau+b\tau^2+c\tau^3+\cdots$ as $\tau\to 0 \Rightarrow$ 代入换元式子求系数 $\Rightarrow dt=(1+2b\tau+3c\tau^2+\cdots)d\tau$
2. **Exact Substitution**: 由 1 的基础, 得到 $f(t)$ 在 t_0 处的展开式, 换元令 $f(t)$ 就等于展开式前几项求解.
3. **If $g=g(t,x)$**: $e^{xf(t)}g(t,x)=e^{xf(t)+\ln(g(t,x))}$, 看 $xf(t)+\ln(g(t,x))$ 的最大值, 类似 1 处理.
   - If $t_0=f(x)$, 换元令 $t=f(x)s$.

Method of Steepest Descents: For $I(x)=\int_C e^{xf(z)}g(z)dz$ as $x\to\infty$. f 容易避开 singular point

If z_0 is saddle point: $f'(z_0)=0, f''(z_0)\ne 0$ $f(z)$ Taylor at z_0: $f(z)=f(z_0)+\frac{f''(z_0)}{2}(z-z_0)^2+\cdots$

1. Let $z=z_0+e^{i\theta}y$ (θ 选让 $f''(z_0)e^{2i\theta}$ 为负实数) $\to I(x)\sim\int_{-\infty}^{\infty}e^{x[f(z_0)-\frac{|f''(z_0)|}{2}y^2+\cdots]}g(z_0)e^{i\theta}dy$
2. If f 不容易避开 singular point: **a.** $\mathrm{Im}(f(z))=\mathrm{Im}(f(z_0))$ and $z_0\in C_0$ to find steepest descent path C_0
3. **b.** 画图看哪些 singular points 在 $C\to C_0$ 变形路径上, 计算 residues. **c.** $I=1$ 部分 + $2\pi i\sum$ residues.
4. **More than one saddle point**: i. 如果通过积分区域可以舍弃, 则舍弃 ii. saddle points: z_k, 取能让 $\mathrm{Re}(f(z_k))$ 最小的.
   iii. 如果多个 saddle points 都满足 $\mathrm{Re}(f(z_k))$ 最小, 则都要计算后叠加.
5. **Exact Substitution**: 由 1 的基础, 得到 $f(z)$ 在 z_0 处的展开式, 换元令 $f(z)$ 就等于展开式前几项求解.
6. **Remark**: 类似 Laplace 3 一样, 可以对 $xf(x)$ 一个整体处理, 必要时 $z_0=f(x)\to$ 换元 $z=f(x)s$.

Residue: If z_0 pole of order m of f, $\Rightarrow \mathrm{Res}(f,z_0)=\frac{1}{(m-1)!}\lim\limits_{z\to z_0}\frac{d^{m-1}}{dz^{m-1}}[(z-z_0)^m f(z)]$

Method of Stationary Phase: $\cos(xf(\theta))\to\mathrm{Re}(e^{ixf(\theta)})$ 回到 Steepest Descents, 同理 $\sin(xf(\theta))$

3 Perturbation Methods in ODEs

Regular Perturbation: For $L(y,\varepsilon)=0$, let $y(x,\varepsilon)=y_0(x)+\varepsilon y_1(x)+\varepsilon^2 y_2(x)+\cdots$

$\Rightarrow$ **a.** 代入 $\mathcal{O}(\varepsilon^n)\to$ **b.** Boundary/Initial Condition 求解 $y_n(x)\to$ **a.**

Singular Perturbation: e.g. $\varepsilon y''+p(x)y'+q(x)y=0$, $y(a)=A$ $y(b)=B$, $a<b$, 大致步骤:

1. Assume boundary layer at $x=a|b$. Let $x=a+\delta(\varepsilon)X$ or $x=b-\delta(\varepsilon)X$
2. Assume inner $Y(X)=y(x)$, chain rule 求导代入原 ODE $\to$ Domain balance 找出 $\delta(\varepsilon)=\varepsilon^{\alpha}$
   Recall PDB: **a.** 至少有两项阶数相同且最低; **b.** 其他项阶数均更高.
3. Assume $Y=Y_0(X)+\varepsilon^{\beta}Y_1(X)+\varepsilon^{2\beta}Y_2(X)+\cdots$ 代入求解 $\to$ 边界/初始条件求解 $Y_0(X)$
4. **Matching**: Consider $X\to\infty|x\to a|b$, If limits not exist (除非 X) $\to$ boundary layer 在另一个端点 $\to 1$
   If limits exist, boundary layer 正确 $\to$ 渐进 outer (x), 算 inner 极限 (X); 算不了保留 $X\to x$ 和 X 的关系代入 $\to$ 比较得系数.

Remark 1: Outer solution 只能用在 away from boundary layer 的区域 (包括边界点).

Remark 2: Inner solution 只能用在 boundary layer 附近的区域 (包括边界点).

Useful Lemma: For $\varepsilon y''+p(x)y'+q(x)y=0$, $y(a)=A$ $y(b)=B$, $a<b$, $p(x)\ne 0$ in $[a,b]$ then
Has boundary layer at $x=a$ if $p>0$; Has boundary layer at $x=b$ if $p<0$.

4 WKB Theory & Solving PDE

WKB (I): For $u''+k^2q(x)u=0$, $k\gg 1$ we have: $u\sim A(x)e^{ik\phi(x)}$. Substitution can get: (q 里不能带 k)

If $q(x)<0$: $u(x)\sim A_+|q|^{-\frac{1}{4}}e^{k\int_{x_0}^{x}\sqrt{|q(t)|}dt}+A_-|q|^{-\frac{1}{4}}e^{-k\int_{x_0}^{x}\sqrt{|q(t)|}dt}$ **(Exp. Decay/Growth)**

If $q(x)>0$: $u(x)\sim A_+q^{-\frac{1}{4}}e^{ik\int_{x_0}^{x}\sqrt{q(t)}dt}+A_-q^{-\frac{1}{4}}e^{-ik\int_{x_0}^{x}\sqrt{q(t)}dt}$ **(Oscillation)**

Turning Point: If $q(x_0)=0$, then x_0 is a WKB turning point. **ps:** 如果 $q(x)$ 含 k, 对其展开保留到 k^0 项

WKB (II): For $u''+k^2q(x)u=0$, $k\gg 1$ Then:

If $q(x)<0$: Assume $u\sim A(x)e^{k\phi(x)}$ then: $u'=(A'+k\phi'A)e^{k\phi}$ and
$u''=(A''+2k\phi'A'+k\phi''A+k^2\phi'^2A)e^{k\phi}$ 代入并比较 $\mathcal{O}(k^2),\mathcal{O}(k)$ (No $\mathcal{O}(1)$)

If $q(x)>0$: Assume $u\sim A(x)e^{ik\phi(x)}$ then: $u'=(A'+ik\phi'A)e^{ik\phi}$ and
$u''=(A''+2ik\phi'A+k\phi''A-k^2\phi'^2A)e^{ik\phi}$ 代入并比较 $\mathcal{O}(k^2),\mathcal{O}(k)$ (No $\mathcal{O}(1)$)

Method of Characteristics|IC: For $u_t+c(x,t)u_x=r(x,t,u)$ with IC $u(x,t_0)=f(x)$:

1. In characteristic form: $\frac{du}{dt}=r(x,t,u)$ on $\frac{dx}{dt}=c(x,t)$ $\to$ 解 ODEs$\to u,x$ 各产生一个未知常数 A,B.
2. Assume characteristic curve (solution of $\frac{dx}{dt}=c(x,t)$) $x(t=t_0)=\xi$ SO $u(t=t_0)=f(\xi)$:
   $\to$ get $u(t=t_0)=f(\xi)$ on $x(t=t_0)=\xi$ $\to$ solve A,B.
3. Substitute A,B back to get u.

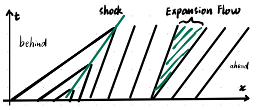
Method of Characteristics|BC: For $u_t+c(x,t)u_x=r(x,t,u)$ with BC $u(x_0,t)=f(t)$:

1. In characteristic form: $\frac{du}{dt}=r(x,t,u)$ on $\frac{dx}{dt}=c(x,t)$ $\to$ 解 ODEs$\to u,t$ 各产生一个未知常数 A,B.
2. Assume characteristic curve (solution of $\frac{dx}{dt}=c(x,t)$) $t(x=x_0)=\xi$ SO $u(x=x_0)=f(\xi)$:
   $\to$ get $u(x=x_0)=f(\xi)$ on $t(x=x_0)=\xi$ $\to$ solve A,B.
3. Substitute A,B back to get u.

Special Cases of Method of Characteristics:

1. 如果 Characteristic curve 和初始曲线/边界条件相交 $\to$ ill-posed & if not unique $\to$ no solution.
2. **For $u_t+c(u)u_x=0$:** If a initial jump $u(x,t_0)=u_L$ for $x<b$ and $u(x,t_0)=u_R$ for $x>b$
   If $c(u_L)<c(u_R)\to$ expansion wave; If $c(u_L)>c(u_R)\to$ shock wave.
3. **For $u_t+c(u)u_x=0, u(x,t_0)=f(x)$:** The **breaking time** $t_b=t_0+\min_{\xi}\frac{-1}{F'(\xi)}$ where $F(\xi)=c(f(\xi))$
   Break time 是 $\frac{du}{dx}=\infty$ (i.e. 分母为 0) 时的最早时间. (通过 chain rule 得到) || **If** $\min_{\xi}F'(\xi)<0\to$ **shock**

Expansion: For $u_t+c(u)u_x=0, u(x,t_0)=u_L|u_R$



1. 正常解 $\to\xi$ 边界值 $\xi_0\to$ 特征曲线 $x(t;\xi)$ 在 ξ_0 边界情况:
   $x=c(u_L)(t-t_0)+\xi_0$ 和 $x=c(u_R)(t-t_0)+\xi_0$
2. $\Rightarrow\frac{x-\xi_0}{t-t_0}=c(u)$ for $c(u_L)(t-t_0)+\xi_0<x<c(u_R)(t-t_0)+\xi_0$
   $\Rightarrow u=c^{-1}[\frac{x-\xi_0}{t-t_0}]$

Conservation Law: $u_t+c(u)u_x=0$: $u_t+[\int c(u)du]_x=0$

Shock Wave|u_L,u_R: For $u_t+c(u)u_x=0, u(x,t_0)=u_L$ for $x<b$ and $u(x,t_0)=u_R$ for $x>b$.

1. 正常解 $\to$ In conservation form: $u_t+[\int c(u)du]_x=0$
   shock velocity $\frac{ds}{dt}=\dot{s}=\frac{[q]}{[\rho]}$ where $q=\int c(u)du$ and $\rho=u$; $[u]:=u_{\text{ahead}}-u_{\text{behind}}$.
   其中: $u_{\text{ahead}}=u_R$, $u_{\text{behind}}=u_L$
2. Solve $\dot{s}$. As shock starts at $x=b$ at $t=t_0$, $s(t=t_0)=b$ get unknown constant.
3. Then solution $u=u_L$ for $x<s(t)$ and $u=u_R$ for $x>s(t)$.

Shock Wave|$f(x)$: For $u_t+c(u)u_x=0, u(x,t_0)=f(x)$ with $t_b=t_0+\min_{\xi}\frac{-1}{F'(\xi)}$ where $F(\xi)=c(f(\xi))$.

1. 正常解 $\to$ In conservation form: $u_t+[\int c(u)du]_x=0$
   shock velocity $\frac{ds}{dt}=\dot{s}=\frac{[q]}{[\rho]}$ where $q=\int c(u)du$ and $\rho=u$; $[u]:=u_{\text{ahead}}-u_{\text{behind}}$.
   其中: $u_{\text{ahead}}=\lim\limits_{x\to s(t)^+}u(x,t)$, $u_{\text{behind}}=\lim\limits_{x\to s(t)^-}u(x,t)$ (which is calculated by characteristics method)
2. Solve $\dot{s}$. As shock starts at $x(t=t_b)$ at $t=t_b$, $s(t=t_b)=x(t=t_b)$ get unknown constant.
   ps: i.e. $x(t=t_b)=\xi_\star+F(\xi_\star)(t_b-t_0)$, 其中 $\xi_\star$ 是使得 $t_b=t_0+\min_{\xi}\frac{-1}{F'(\xi)}$ 的 ξ.

Remark: If $u_t+c(u)u_x=0$, 大体思路都不变, 除了 In conservation form: $u_t+[\int c(u)du]_x=r(x,t,u)$.

Remark: 可能 Expansion 和 Shock 冲突 $\to$ 原先 Shock 消失并留 *Expansion fan* 合并. $\to$ 寻找第一次 $t_b>0$ 可能发生的时间. $\to$ 解新 $\dot{s}$; u_R, u_L 是 *expansion fan* 值,*shock* 前的值. $\to$ 用原先 $s(t=t_b)$ 解新 s 未知数. $\to$ 新 u, 留 expansion fan.

5 Sturm-Liouville Theory

Regular S-L: $\frac{d}{dx}[p(x)\frac{dy}{dx}]+q(x)y+\lambda\rho(x)y=0$, with $a<x<b$ and BCs:

- $\beta_1y(a)+\beta_2y'(a)=0$ and $\beta_3y(b)+\beta_4y'(b)=0$.

1. **Eigenvalues**: **a.** All λ are *real*; **b.** Infinite λ's: $\lambda_1<\lambda_2<\lambda_3<\cdots$; **c.** 存在最小不存在最大.
2. **Eigenfunctions**: **a.** $\lambda_n\leftrightarrow u_n$ *unique*; **b.** $u_n(x)$ has $n-1$ *zeros in* (a,b); **c.** $u_n\to$ orthogonal basis.
3. **Algebra**: **a.** $\langle f,g\rangle=\int_a^b f(x)g(x)\rho(x)dx$; **b.** Any $f(x)\to f(x)=\sum_{n=1}^{\infty}a_nu_n(x)$, $a_n=\frac{\langle f,u_n\rangle}{\langle u_n,u_n\rangle}$.
4. **Rayleigh Quotient**: $\lambda_n=\frac{[-pu_nu_n']_a^b+\int_a^b(pu_n'^2-qu_n^2)dx}{\int_a^b\rho u_n^2dx}$.
5. **Useful Tools**: u_n,λ_n homogeneous eigenvalues/functions. $\Rightarrow [pu_n']'+qu_n+\lambda_n\rho u_n=0$.

How to Solve: Seperation of variables $\to$ Two ODEs $\to$ 优先解 $y''\pm\lambda y=0$ and $\lambda=\pm\xi^2\to$ eigenvalues/functions.

Then Superposition $\to$ e.g. $u(x,t)=\sum A_nX_n(x)T_n(t)\to$ with Coefficients: IC/BC & $A_n=\frac{\langle f,u_n\rangle}{\langle u_n,u_n\rangle}$.

⋆ For higher order PDEs: e.g. (2D) $\langle f,g\rangle=\int_a^b\int_c^d f(x,y)g(x,y)\rho_1(x)\rho_2(y)dxdy$ ↓(not include t)

- where: **a.** a,b,c,d BCs for x,y. **b.** ρ_1,ρ_2 are weight functions for x,y in their sub-SL problems.

For IC (t), if contain x $\Rightarrow$ 最后使用, 用 orthogonality 求系数.

Non-Homogeneous: 对于一般的**BCs** 非齐次 (RHS$\ne 0$, IC 正常) $\to$ e.g. Heat $T_t=\nu\Delta T$ with BCs.

a. Steady State $T_t=0$ (对于 Heat 来说更好解加的) find $S(x,y)$ 满足 BCs

b. Solve $S(x,y)$ and then $T=S+\Theta$, where Θ 满足齐次 BCs 和 PDE.

Useful Tools: **a.** For $y''+\lambda y=0$ with $y(0)=y(L)=0\to$ Only need discuss $\lambda>0$ else trivial solution.

b. For $y''+\lambda y=0$ with $y'(0)=y'(L)=0\to$ Only need discuss $\lambda\ge 0$ else trivial solution.

c. For $y''+\lambda y=0$ with $y(0)=y'(L)=0\to$ Only need discuss $\lambda>0$ else trivial solution.

Laplace Transformation: $\mathcal{L}\{f(t)\}=\int_0^{\infty}e^{-st}f(t)dt=\overline{f}(s)$ $\mathcal{L}^{-1}\{\overline{f}(s)\}=\frac{1}{2\pi i}\int_{\gamma-i\infty}^{\gamma+i\infty}e^{st}\overline{f}(s)ds$.

a. γ lies to *right of all singularities* of $\overline{f}(s)$.

b. Method: PDE|ICs|BCs 对t做 Laplace $\to$ 解 ODE 如有 δ 分类讨论 $\to$ inverse Laplace

Remark: **Need initial value.**

Fourier Transformation: $\mathcal{F}\{f(x)\}=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}f(x)e^{ikx}dx$ $\mathcal{F}^{-1}\{\overline{f}(k)\}=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\overline{f}(k)e^{-ikx}dk$.

a. We suppose $f(x)\to 0$ as $x\to\pm\infty$ and $\mathcal{F}\{f'(x)\}=-ik\mathcal{F}(f(x))$ $\mathcal{F}\{f^{(n)}(x)\}=(-ik)^n\mathcal{F}(f(x))$.

b. Method: PDE|ICs|BCs 对x做 Fourier $\to$ 解 ODE 如有 δ 分类讨论 $\to$ inverse Fourier

Remark: **Need with infinite space domain.**

Useful Tool when have δ:

1. If ODE order ≥ 2: **continuous** at δ point (LHS=RHS)
2. **Jump condition**: $\int_{\xi^-}^{\xi^+}$ PDE|ODE LHS$dx=\int_{\xi^-}^{\xi^+}$ PDE|ODE RHSdx once we have $\delta(x-\xi)$.

6 Appendix

x, x^2, x^3 **as** $x\to 0$, x^{-1}, x^{-2}, x^{-3} **as** $x\to$ **inf**

6.1 ODE

$y'+Py=Q$: $\mu y=\int Q(x)\mu dx+C$, $\mu(x)=e^{\int P(x)dx}$ $y'=Py$: $y=Ce^{\int P(x)dx}$

$y''+k^2y=0$: $y=A\cos(kx)+B\sin(kx)=Ae^{ikx}+Be^{-ikx}, k>0$

$y''-k^2y=0$: $y=Ae^{kx}+Be^{-kx}, k>0$

$ay''+by'+cy=0$: Characteristic equation: $ar^2+br+c=0$ $r_{1,2}=\mu\pm i\nu$

a. $\Delta>0$: $y=Ae^{r_1x}+Be^{r_2x}$; **b.** $\Delta=0$: $y=(A+Bx)e^{\mu x}$; **c.** $\Delta<0$: $y=e^{\mu x}(A\cos(\nu x)+B\sin(\nu x))$

6.2 Integral

Gaussian Integral: $\int_{-\infty}^{\infty}e^{-ax^2}dx=\frac{\sqrt{\pi}}{\sqrt{a}}$ $\int_0^{\infty}e^{-xt^n}t^m=\frac{1}{n}\Gamma(\frac{m+1}{n})x^{-\frac{m+1}{n}}$

Integral: $\int\tan x=-\ln|\cos x|=\ln|\sec x|$ $\int\cot x=\ln|\sin x|=-\ln|\csc x|$

Techniques: **a.** e.g. $\int e^{at}\cos(bt)dt=\mathrm{Re}(\int e^{(a+ib)t}dt)$

Basis: For $\langle\cdot,\cdot\rangle$ with $\langle f,g\rangle=\int_a^b f(x)g(x)dx$ ($\rho=1$) we have: $\langle\sin(n\pi(x-a)/L),\cos(m\pi(x-b)/L)\rangle=0$

For $\langle\sin(n\pi(x-a)/L),\sin(m\pi(x-b)/L)\rangle=\frac{L}{2}\delta_{nm}$, $\langle\cos(n\pi(x-a)/L),\cos(m\pi(x-b)/L)\rangle=\frac{L}{2}\delta_{nm}$.

6.3 Expansion

For $x\to a$: Taylor at $x=a$: $f(x)\sim f(a)+f'(a)(x-a)+\frac{f''(a)}{2!}(x-a)^2+\cdots$ as $x\to a$

Taylor|$x\to 0$: $f(x+h)\sim f(x)+f'(x)h+\frac{f''(x)}{2!}h^2+\cdots$ as $h\to 0$

$\sin(x)\sim x-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots$ as $x\to 0$ $e^x\sim 1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots$ as $x\to 0$

$\cos(x)\sim 1-\frac{x^2}{2!}+\frac{x^4}{4!}-\cdots$ as $x\to 0$ $\ln(1+x)\sim x-\frac{x^2}{2}+\frac{x^3}{3}-\cdots$ as $x\to 0$

$\sinh(x)\sim x+\frac{x^3}{3!}+\frac{x^5}{5!}+\cdots$ as $x\to 0$ $\cosh(x)\sim 1+\frac{x^2}{2!}+\frac{x^4}{4!}+\cdots$ as $x\to 0$

$\tan(x)\sim x+\frac{x^3}{3}+\frac{2x^5}{15}+\cdots$ as $x\to 0$ $\cot(x)\sim\frac{1}{x}-\frac{x}{3}-\frac{x^3}{45}-\cdots$ as $x\to 0$

For $x\to\infty$: $\arctan(x)\sim\frac{\pi}{2}-\frac{1}{x}+\frac{1}{3x^3}-\cdots$ as $x\to\infty$

$\ln(a+x)=\ln x+\ln(1+\frac{a}{x})\sim\ln x+\frac{a}{x}-\frac{a^2}{2x^2}+\cdots$ as $x\to\infty$

$\frac{1}{(a+x)^k}=\frac{1}{x^k}\cdot\frac{1}{(1+\frac{a}{x})^k}\sim\frac{1}{x^k}-\frac{ka}{x^{k+1}}+\frac{k(k+1)a^2}{2x^{k+2}}-\cdots$ as $x\to\infty$

$\frac{1}{(a+x)^k}=\frac{1}{a^k}\cdot\frac{1}{(1+\frac{x}{a})^k}\sim\frac{1}{a^k}-\frac{kx}{a^{k+1}}+\frac{k(k+1)x^2}{2a^{k+2}}+\cdots$ as $x\to 0$

sin|cos: $a\sin(x)+b\cos(x)=\sqrt{a^2+b^2}\sin(x+\varphi)$ $a\cos(x)+b\sin(x)=\sqrt{a^2+b^2}\cos(x-\varphi)$, φ 过 (a,b).

- $\sin(x)+\sin(y)=2\sin(\frac{x+y}{2})\cos(\frac{x-y}{2})$ | $\sin(x)\cos(y)=\frac{1}{2}[\sin(x+y)+\sin(x-y)]$ s+s=sc,s-s=cs,c+c=cc,c-c=-ss

Inequalities: $\ln(x+1)<x$, $e^x>x+1$, $(1+x)^a<e^{ax}$

6.4 Other

Nabla: $\nabla^2u=u_{xx}+u_{yy}+u_{zz}+\cdots$ $\nabla\cdot\mathbf{F}=\frac{\partial F_x}{\partial x}+\frac{\partial F_y}{\partial y}+\frac{\partial F_z}{\partial z}$

Binomial Expansion: $(a+b)^n=\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k$ $\binom{n}{k}=\frac{n!}{k!(n-k)!}$ $(1+x)^n=\sum_{k=0}^{n}\binom{n}{k}x^k$

Sin/Cos: $\sin[\frac{2n+1}{2}\pi]=(-1)^n$ $\cos(n\pi)=(-1)^n$

7 Green's Function

ps: To solve *forced equation*. 即 PDE RHS $\neq 0$

Green's Function: For PDE $Lu(x,t) = F(x,t)$ where L is *linear* differential operation with BCs and ICs.

1. Consider $G(x,t;\xi,\tau)$ satisfying $LG(x,t;\xi,\tau) = \delta(x-\xi)\delta(t-\tau)$ with BCs and ICs.
2. Solve $G \to G(x,t;\xi,\tau)$.
3. Then $u(x,t) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} G(x,t;\xi,\tau)F(\xi,\tau)d\xi d\tau$.

ps: 可以拓展到更多的空间维度, e.g. $G(x,y,t;\xi,\eta,\tau) \to LG(x,y,t;\xi,\eta,\tau) = \delta(x-\xi)\delta(y-\eta)\delta(t-\tau)$

- How to solve: $LG(x,t;\xi,\tau) = \delta(x-\xi)\delta(t-\tau)$ with BCs and ICs. ↓

a. Fourier Transform (infinite space domain (x,t), [2 dimensions])

1. Take Fourier transform in x → Get ODE → Solve ODE with $t \neq \tau$. → Get $\overline{G}$ split by $t < \tau$ and $t > \tau$.
2. ICs → Get some unknown constants in $\overline{G}$.
3. Apply **Jump Condition** to *ODE*, $\int_{\tau-}^{\tau+} \cdot \, dt$ → Get unknown constant.
4. Inverse Fourier → $G(x,t;\xi,\tau)$.

b. Separation of Variables (finite space domain (x))

1. Need homogeneous BCs → Assume $t \neq \tau$ **and** $x \neq \xi$ → Seperation of variables → Get λ_n and $X_n(x)$.
2. Assume $G = \sum_n g_n(t) X_n(x)$ → Substitute into PDE $LG = \delta(x-\xi)\delta(t-\tau)$.
3. Use orthogonality of X_n → Get $\delta(x-\xi) = \sum_n d_n X_n(x)$ and find d_n.
4. 通过前面 substitute 得到的对比 terms → Get ODE for $g_n(t)$ → Solve ODE split by $t < \tau$ and $t > \tau$.
5. ICs → Get some unknown constants in $g_n(t)$. → **Continuity** at $t = \tau$ → some unknown constants.
6. Apply **Jump Condition** to *ODE*, $\int_{\tau-}^{\tau+} \cdot \, dt$ → Get unknown constant → Get $g_n(t)$ and we solve it :)

Remark: 可以拓展为更多空间维度 → Separation of variables 后留空间不算到最后假设来找.

e.g. $G_t - \Delta G = \delta(x-\xi)\delta(y-\eta)\delta(t-\tau) \to \cdots \to G = \sum_{m,n} g_{m,n}(t)X_n(x)Y_m(y) \to \cdots$

c. High Variables / Laplace Transform

Remark: 如 $F(x,t)$ 给具体函数 → 考虑齐次 PDE → Eigenfunction $X_n(x) \to u = \sum_n g_n(t)X_n(x)$
→ RHS 含 x 的写为 $\sum_n d_n X_n(x)$ → 比较系数 → 得到 $g_n(t)$ 的 ODE → Solve ODE → 最后得到 $u(x,t)$
i.e. 跳过求 G 的步骤, 直接用 X_n 展开 RHS

8 Laplace Transform Table

$f(t)$	$\mathcal{L}\{f(t)\}$	$f(t)$	$\mathcal{L}\{f(t)\}$
1	$\frac{1}{s}$	t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$	$t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}}$
$\sin(at)$	$\frac{a}{s^2+a^2}$	$t^n \sin(at)$	$\frac{n!a}{(s^2+a^2)^{n+1}}$
$\cos(at)$	$\frac{s}{s^2+a^2}$	$t^n \cos(at)$	$\frac{n!s}{(s^2+a^2)^{n+1}}$
$\sinh(at)$	$\frac{a}{s^2-a^2}$	$t^n \sinh(at)$	$\frac{n!a}{(s^2-a^2)^{n+1}}$
$\cosh(at)$	$\frac{s}{s^2-a^2}$	$t^n \cosh(at)$	$\frac{n!s}{(s^2-a^2)^{n+1}}$
$u_b(t)$	$\frac{e^{-bs}}{s}$	$\delta(t-b)$	e^{-bs}

$f(t)$	$\mathcal{L}\{f(t)\}$
$f'(t)$	$s\overline{f}(s) - f(0)$
$\int_0^t f(\tau)d\tau$	$\frac{1}{s}\overline{f}(s)$
$e^{at}f(t)$	**s-shift** $\overline{f}(s-a)$
$f(t-a)u_a(t)$	**t-shift** $e^{-as}\overline{f}(s)$
$\int_0^t f(t-\tau)g(\tau)d\tau$	**Convolution** $\overline{f}(s)\cdot\overline{g}(s)$

$f^n(t)$: $s^n\overline{f}(s) - s^{n-1}f(0) - \cdots - f^{(n-1)}(0)$

$f * g = \int_0^t f(t-\tau)g(\tau)d\tau$

ps: $u_b(t) = H(t-b) = \begin{cases} 0, & t<b \\ 1, & t\geq b \end{cases}$ $\int_{-\infty}^{\infty} \delta(t-b)\phi(t)dt = \int_{b-}^{b+} \delta(t-b)\phi(t)dt = \phi(b)$

Hyperbolic Functions: $\sinh(x) = \frac{e^x - e^{-x}}{2}$ $\cosh(x) = \frac{e^x + e^{-x}}{2}$

$1 + \tanh^2(x) = \text{sech}^2(x)$ $\coth^2(x) - 1 = \text{csch}^2(x)$

ps: $1 + \tan^2(x) = \sec^2(x)$ $1 + \cot^2(x) = \csc^2(x)$

9 Appendix II

Useful Tools: $\int_0^L \sin^2(ax)dx = \frac{L}{2} - \frac{\sin(2aL)}{4a}$ $\int_0^L \cos^2(ax)dx = \frac{L}{2} + \frac{\sin(2aL)}{4a}$

$\int_0^L \sin(ax)\cos(ax)dx = \frac{\sin^2(aL)}{2a}$

Quick Calculations: $\int xe^{ax}dx = \frac{1}{a}xe^{ax} - \frac{1}{a^2}e^{ax}$ $\int x^2e^{ax}dx = \frac{1}{a}e^{ax}x^2 - \frac{2}{a^2}e^{ax}x + \frac{2}{a^3}e^{ax}$

If $y_p = P(x)\sin(\beta x) + Q(x)\cos(\beta x)$ where P, Q are polynomials, then

· $y_p' = P'\sin(\beta x) + Q'\cos(\beta x) + \beta P\cos(\beta x) - \beta Q\sin(\beta x)$ and

· $y_p'' = P''\sin(\beta x) + Q''\cos(\beta x) + 2\beta P'\cos(\beta x) - 2\beta Q'\sin(\beta x) - \beta^2 P\sin(\beta x) - \beta^2 Q\cos(\beta x)$

$y_p'' + \beta^2 y_p = P''\sin(\beta x) + Q''\cos(\beta x) + 2\beta P'\cos(\beta x) - 2\beta Q'\sin(\beta x)$

$y_p'' - \beta^2 y_p = (P'' - 2\beta^2 P)\sin(\beta x) + (Q'' - 2\beta^2 Q)\cos(\beta x) + 2\beta P'\cos(\beta x) - 2\beta Q'\sin(\beta x)$

Note for $ay'' + by' + cy = f(x)$: If $r = \alpha \pm i\beta$, and $f(x)$ contain $e^{\alpha x}\sin(\beta x)|\cos(\beta x)$ then

· 一个根: $\times x$; 两个根: $\times x^2$. ps: if $r = 0$ remember $e^{0x} = 1$